



UNIVERSITY OF
WATERLOO

MTE 203 – Lecture 9

Midterm Exam Fall 2016 – Problem 2

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A curve of intersection is defined by the surfaces given below, for the values in the positive octant only.

$$x^2 + y^2 + z^2 = 4, \quad x + z = 2$$

- a. Sketch the curve of intersection
- b. Find a vector valued function for the curve directed so that z decreases from 2 to 0 using the parametrization,

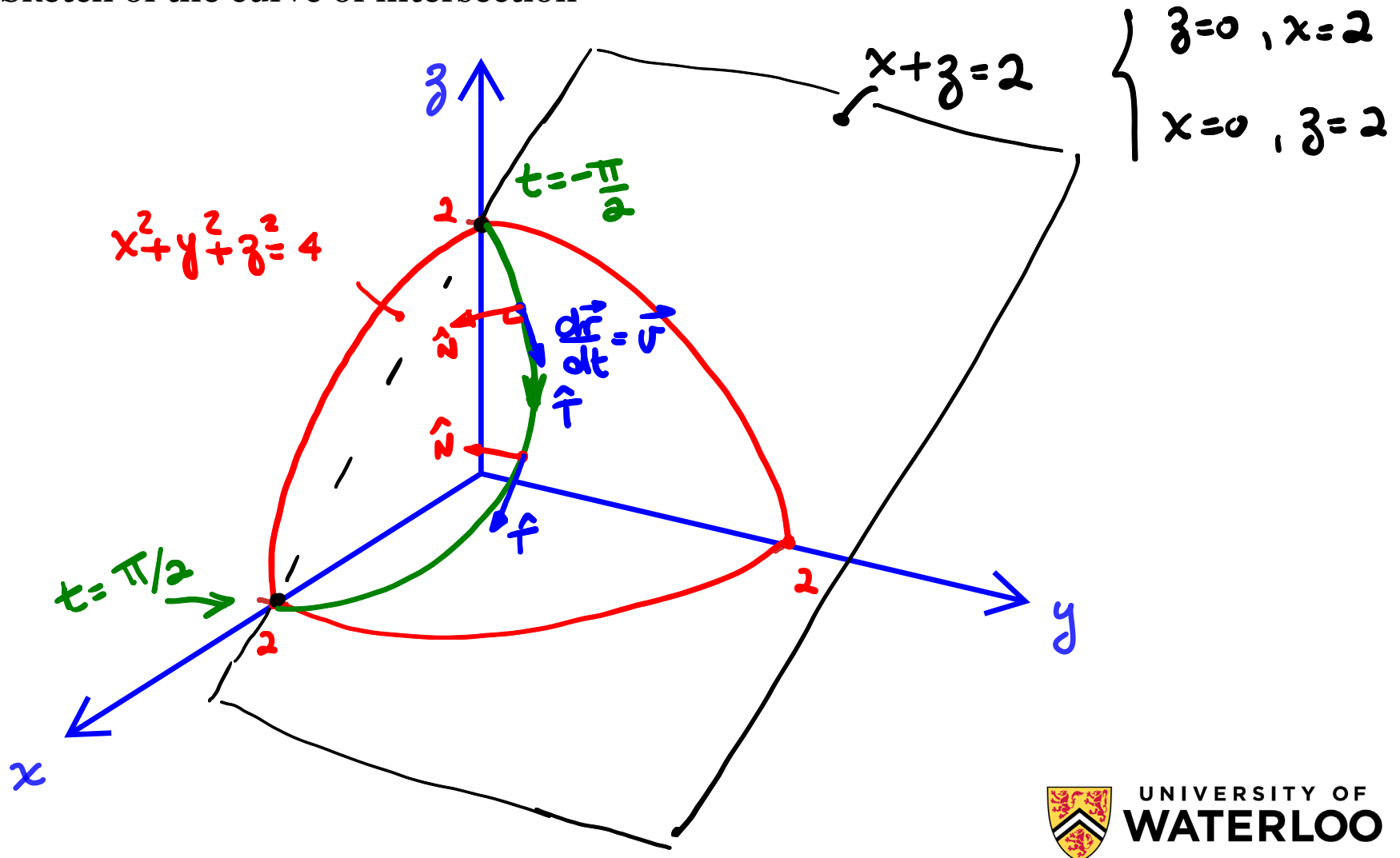
$$x = 1 + \sin t$$

- c. What is the range of the parameter t for the curve of intersection? Clearly indicate the direction of the curve in your sketch.
- d. Calculate the unit tangent vector $\hat{T}$ in the direction of increasing t .
- e. Find the length of the curve obtained in parts b and c
- f. Without calculating the moving right-handed Frenet vector Frame, can you determine the equation of the osculating plane at the point $(1, \sqrt{2}, 1)$ in the curve?



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a. Sketch of the curve of intersection



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- b. Find a vector valued function for the curve directed so that z decreases from 2 to 0 using the parametrization,

$$x = 1 + \sin t$$

FROM PLANE: $x + z = 2 \Rightarrow z = 1 - \sin t$

FROM SPHERE: $x^2 + y^2 + z^2 = 4$

$$(1 + \sin t)^2 + y^2 + (1 - \sin t)^2 = 4$$

$$\vdots$$
$$y = 2 - 2\sin^2 t = 2\cos^2 t$$

$$y = \pm \sqrt{2} \cos t$$



POSITIVE

OCTANT



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THE VECTOR FUNCTION ,

$$\vec{r}(t) = (1 + \sin t)\hat{i} + \sqrt{2} \cos t \hat{j} + (1 - \sin t)\hat{k}$$

- c. What is the range of the parameter t for the curve of intersection? Clearly indicate the direction of the curve in your sketch.
-

$$\text{At } z=2 \rightarrow \underbrace{1 - \sin t}_z = 2 \rightarrow \sin t = -1, t = -\pi/2$$

$(0, 0, 2)$

$$\text{At } z=0 \rightarrow \underbrace{1 + \sin t}_x = 2 \rightarrow \sin t = 1, t = \pi/2$$

$(2, 0, 0)$

$$-\pi/2 \leq t \leq +\pi/2$$



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d. Calculate the unit tangent vector $\hat{T}$ in the direction of increasing t .

$$\hat{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \cos t \hat{i} - \sqrt{2} \sin t \hat{j} - \cos t \hat{k}$$

$$|\vec{v}(t)| = \sqrt{\cos^2 t + 2 \sin^2 t + \cos^2 t} = \sqrt{2}$$

$2(\cos^2 t + \sin^2 t)$

$$\hat{T} = \frac{\sqrt{2}}{2} \cos t \hat{i} - \sin t \hat{j} - \frac{\sqrt{2}}{2} \cos t \hat{k}$$



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e. Find the length of the curve obtained in parts b and c

$$\begin{aligned} L &= \int_{-\pi/2}^{\pi/2} |\vec{v}(\tau)| d\tau \\ &= \int_{-\pi/2}^{\pi/2} \sqrt{2} dt = \sqrt{2} \Big|_{-\pi/2}^{\pi/2} \\ L &= \sqrt{2} (\pi) \rightarrow \end{aligned}$$



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- f. Without calculating the moving right-handed Frenet vector Frame, can you determine the equation of the osculating plane at the point $(1, \sqrt{2}, 1)$ in the curve?

THE CURVE C LIES IN ' $x+z=2$ ' AND SO

ALL $\hat{T}$ 'S AND $\hat{N}$ 'S

OSCULATING PLANE IS THAT PLANE CONTAINING
 $\hat{T}$ AND $\hat{N}$ AT $(1, \sqrt{2}, 1)$

$$x+z=2$$

